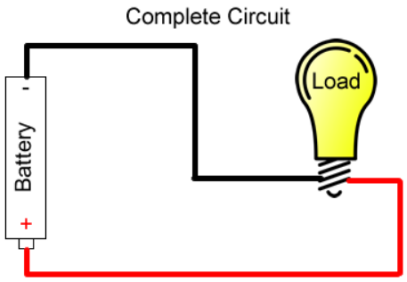
Module -I

Basic Concepts:

Introduction:

An electric circuit is an interconnection of electrical elements.

A simple electric circuit is shown below.



- ★ An electric circuit has three parts: An energy source – like a battery or mains power, an energy receiver – like a lightbulb and an energy pathway – like a wire.
- ★ Such a simple circuit can exist by itself with several applications, such as a flashlight, a searchlight, and so forth.
- ★ The bulb will glow only if the circuit has a closed path for the flow of electrons.
- ★ The direction of flow of conventional current is from positive to negative terminal.

Systems of Units:

All measurable quantities are communicated in a standard language that virtually all professionals can understand, irrespective of the country in which the measurement is conducted. Such an international measurement language is the **International System of Units (SI)**, adopted by the General Conference on Weights and Measures in 1960.

Six basic SI units and one derived unit relevant to this text.

Quantity	Basic unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Luminous intensity	candela	cd
Charge	coulomb	C

One great advantage of the SI unit is that it uses prefixes based on the power of 10 to relate larger and smaller units to the basic unit.

The SI prefixes.

Multiplier	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{2}	hecto	h
10	deka	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a

Charge and Current:

Most basic quantity in an electric circuit is the electric charge.

Charge is an electrical property of the atomic particles of which matter consists, measured in coulombs (C).

- In 1 C of charge, there are 1/(1.602 × 10^−19) = 6.24 × 10^18 electrons.
- 1 electronic charge, e = 1.602 × 10^−19 C.
- The law of conservation of charge states that charge can neither be created nor destroyed, only transferred. Thus, the algebraic sum of the electric charges in a system does not change.

Electricity - a form of energy that results from the flow of charged particles.

Electric current - the time rate of change of charge, measured in amperes (A).

Mathematically, given by

$$i \triangleq \frac{dq}{dt}$$

1 ampere = 1 coulomb/second

- Benjamin Franklin - an American scientist and inventor who introduced current flow as the movement of positive charges.
- Andre-Marie Ampere - a French mathematician and physicist who defined the electric current and developed a way to measure it in the 1820s; thus, laid the foundation of electrodynamics.
- Alessandro Antonio Volta - an Italian physicist, invented the electric battery, which provided the first continuous flow of electricity, and the capacitor.

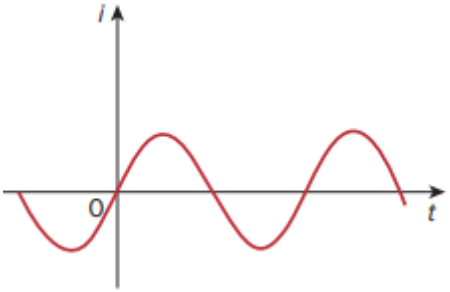
Types of current:

Based on whether these currents can be constant or time varying.



Direct current (dc) - a current that remains constant with time; flows in one direction.

Eg: Any electronic device with a battery for a power source. Rechargeable devices like laptops and cell phones come with an AC adapter that converts alternating current to direct current.



Alternating current (ac) - a current that varies sinusoidally with time.

Eg: of alternating current (ac): current you use in your house to run the air conditioner , refrigerator , washing machine, and other electric appliances.

Voltage and Power:

Voltage (or potential difference) - the energy required to move a unit charge through an element, measured in volts (V).

Power - the time rate of expending or absorbing energy, measured in watts (W).

$$p = vi$$

The law of conservation of energy states that energy can neither be created nor be destroyed. Although, it may be transformed from one form to another. Therefore,

+Power absorbed = −Power supplied

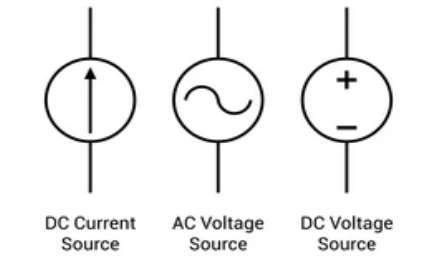
The algebraic sum of power in a circuit, at any instant of time, must be zero: $\sum p = 0$

Circuit Elements:

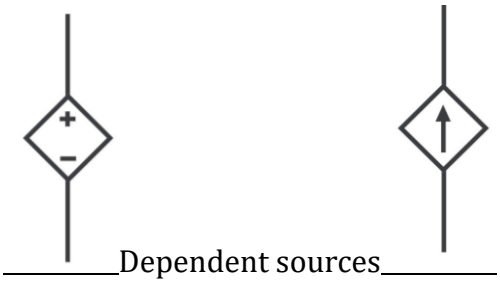
It is an individual electrical component like R, C, L and sources like voltage and current connected between two terminals.

Examples:





Independent sources



Dependent sources

2 types of elements found in electric circuits:

1. Passive elements
2. Active elements.

An active element is capable of generating energy while a passive element is not. Examples of passive elements are resistors, capacitors, transforms and inductors.

Examples of active elements include generators, batteries, transistors and operational amplifiers.

- The most important active elements are voltage or current sources that generally deliver power to the circuit connected to them.

There are 2 kinds of sources:

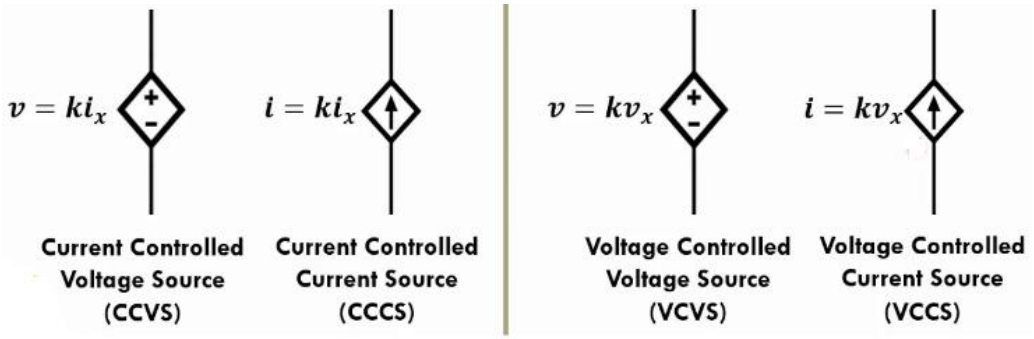
1. Independent sources
2. Dependent sources.

An ideal independent source is an active element that provides a specified voltage or current that is completely independent of other circuit elements.

An ideal dependent (or controlled) source is an active element in which the source quantity is controlled by another voltage or current.

Dependent sources are usually designated by diamond-shaped symbols. Since the control of the dependent source is achieved by a voltage or current of some other element in the circuit, and the source can be voltage or current, there are 4 types of dependent sources, namely:

1. A voltage-controlled voltage source (VCVS).
2. A current-controlled voltage source (CCVS).
3. A voltage-controlled current source (VCCS).
4. A current-controlled current source (CCCS).



OHM'S LAW

Ohm's law states that the voltage v across a resistor is directly proportional to the current i flowing through the resistor.

$$v \propto i$$

Or

$$v = iR$$

The resistance R of an element denotes its ability to resist the flow of electric current; it is measured in ohms (Ω).

$$R = \frac{v}{i}$$

A short circuit is a circuit element with resistance approaching zero.

An open circuit is a circuit element with resistance approaching infinity.

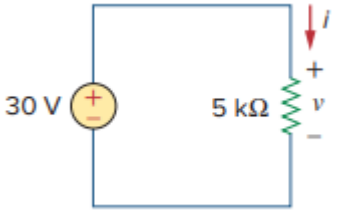
Conductance is the ability of an element to conduct electric current; it is measured in mhos (℧) or siemens (S).

$$G = \frac{1}{R} = \frac{i}{v}$$

Solved Ex1: An electric iron draws 2 A at 120 V. Find its resistance.

From Ohm's law, R = v /i =120/ 2 = 60 Ω

Solved Ex2: In the circuit shown in Fig., calculate the current i, the conductance G, and the power p.



$$i = \frac{v}{R} = \frac{30}{5 \times 10^3} = 6 \text{ mA}$$

The conductance is

$$G = \frac{1}{R} = \frac{1}{5 \times 10^3} = 0.2 \text{ mS}$$

$$p = vi = 30(6 \times 10^{-3}) = 180 \text{ mW}$$

Gustav Robert Kirchhoff - a German physicist, stated two basic laws in 1847 concerning the relationship between the currents and voltages in an electrical network. Kirchhoff's laws, along with Ohm's law, form the basis of circuit theory.

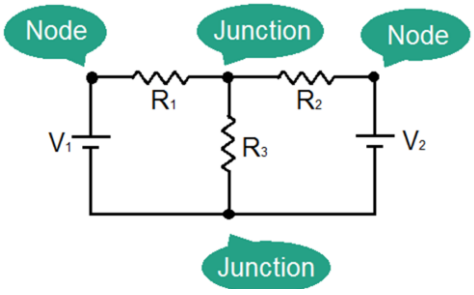
Nodes, Branches, and Loops:

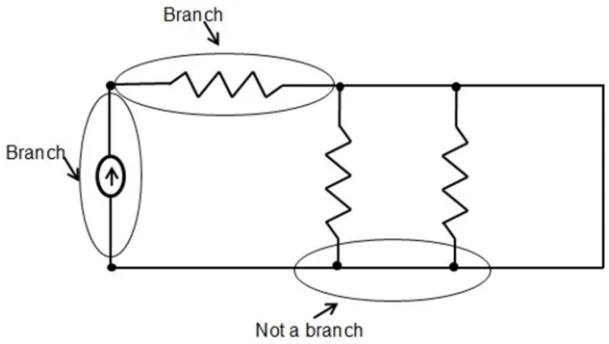
A **branch** represents Portion of circuit between 2 nodes which can deliver or absorb energy. It meets /stays between junctions.

A **node** is the point of connection between two or more branches.

Junction: Point where three or more elements meet

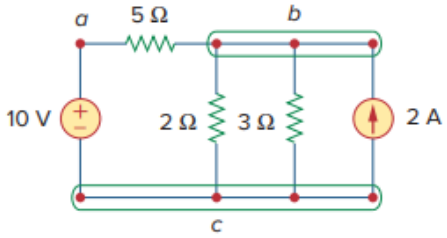
If a short circuit (a connecting wire) connects two nodes, the two nodes constitute a single node.





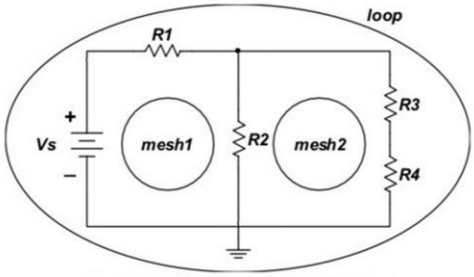
Note: *Short Circuit between 2 nodes is not a branch.*

Solved Example 1 : Find the number of branches, nodes in the given circuit.



Circuit has five branches, three nodes a, b, and c.

A **loop** is any closed path in a circuit.
A **Mesh** is an Elementary part of the loop/ Path that does not contain any other path.

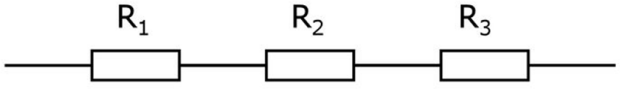


Note: A mesh can stay within the loop but the loop cannot stay within the mesh.

A network with b branches, n nodes, and *l* independent loops(mesh) will satisfy the **fundamental theorem of network topology:**

$$\mathbf{b = l + n - 1}$$

Series: Two or more elements are in series if they exclusively share a single node and consequently carry the same current.



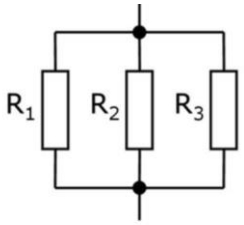
$$R_T = \sum_n R_n$$

Resistors in series: $R_{Total}=R_1+R_2+R_3...$

The equivalent conductance of resistors connected in series is

$$\frac{1}{G_{eq}} = \frac{1}{G_1} + \frac{1}{G_2} + \frac{1}{G_3} + \cdots + \frac{1}{G_N}$$

Parallel: Two or more elements are in parallel if they are connected to the same two nodes and consequently have the same voltage across them.



Resistors in parallel

$$\frac{1}{R_T} = \sum_n \frac{1}{R_n}$$

$$= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \cdots$$

Two parallel resistors:

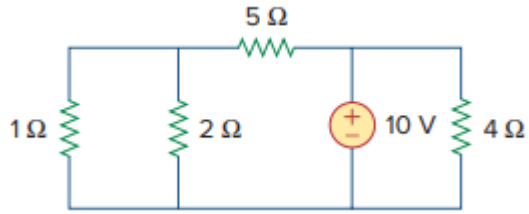


$$R_T = \frac{R_1 R_2}{R_1 + R_2}$$

The equivalent conductance of resistors connected in parallel is the sum of their individual conductances.

$$G_{eq} = G_1 + G_2 + G_3 + \cdots + G_N$$

Solved Ex 2: Determine the number of branches, nodes, junction, meshes and loops in the circuit shown in Fig. below. Identify which elements are in series and which are in parallel. Also check whether the number of network elements is equal to the number of branches present.

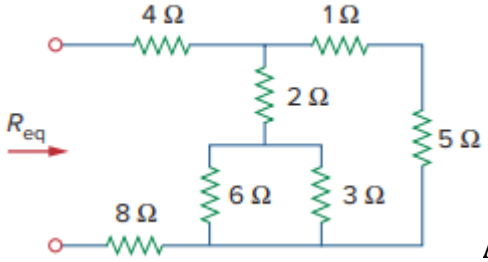


Ans: from the figure shown

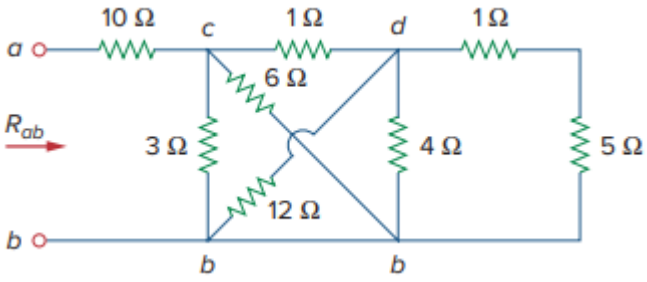
- ➔ 5 branches, 3 nodes, 3 meshes and 6 loops
- ➔ The 1-Ω and 2-Ω resistors are in parallel. The 4-Ω resistor and 10-V source are also in parallel.
- ➔ Fundamental theorem , b = *l* + n − 1
  - ◆ b= 5, n=3 gives *l*=3 (no of meshes).
- ➔ No of network elements = 5= no of branches

Exercise:

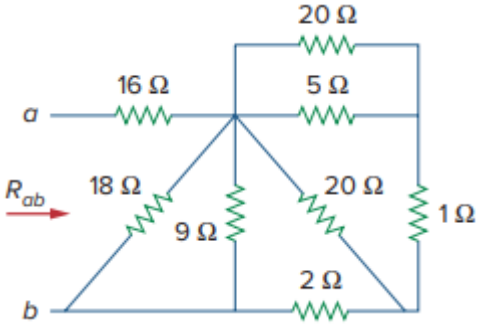
Find equivalent resistance for the following circuits:

1. 

Ans: 14.4Ω

2. 

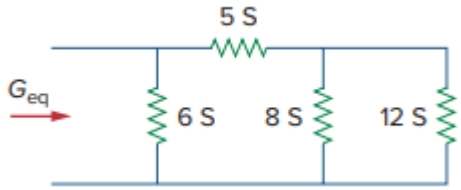
Ans: 11.2Ω

3. 

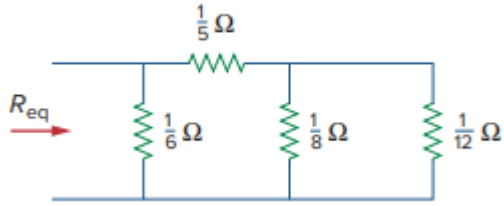
Ans: 19Ω

Solved Example:

Find the equivalent conductance Geq for the circuit shown:



Hint: First Convert into Resistance



$$R_{eq} = \frac{1}{6} \Big\| \left(\frac{1}{5} + \frac{1}{8} \Big\| \frac{1}{12} \right) = \frac{1}{6} \Big\| \left(\frac{1}{5} + \frac{1}{20} \right) = \frac{1}{6} \Big\| \frac{1}{4}$$

$$= \frac{\frac{1}{6} \times \frac{1}{4}}{\frac{1}{6} + \frac{1}{4}} = \frac{1}{10}\,\Omega$$

$$G_{eq} = \frac{1}{R_{eq}} = 10\ \text{S}$$

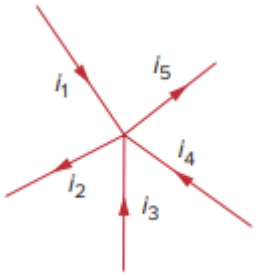
Kirchhoff's Laws:

Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.

Mathematically, KCL implies that

$$\sum_{n=1}^{N} i_n = 0$$

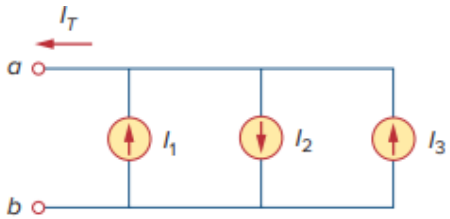
where N is the number of branches connected to the node and i_n is the nth current entering (or leaving) the node. For the figure shown



$$i_1 + i_3 + i_4 = i_2 + i_5$$

The sum of the currents entering a node is equal to the sum of the currents leaving the node.

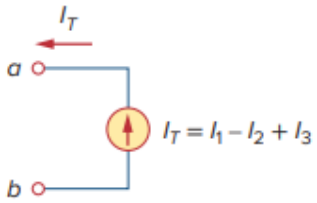
Currents in Parallel:



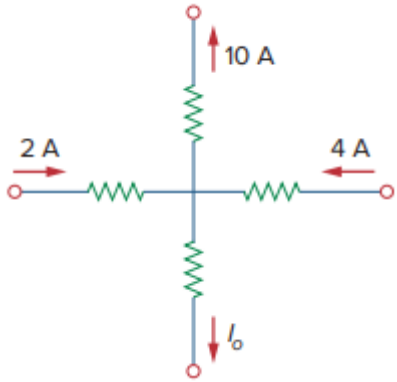
Ex1: Applying KCL;

$I_T + I2 = I1 + I3$

or $IT = I1 - I2 + I3$



Exercise Qsn: Find the current Io

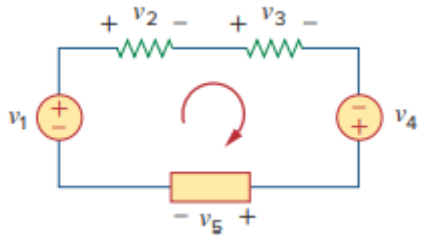


Ans: Io= 2+4-10=-4A

Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero. Expressed mathematically, KVL states that

$$\sum_{m=1}^{M} v_m = 0$$

where M is the number of voltages in the loop (or the number of branches in the loop) and Vm is the mth voltage



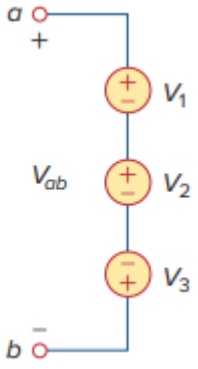
$-v1 + v2 + v3 - v4 + v5 = 0$

Rearranging terms gives $v2 + v3 + v5 = v1 + v4$

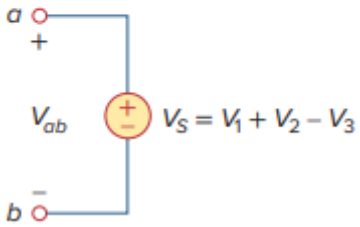
Sum of voltage drops = Sum of voltage rises

Voltages in Series:

When voltage sources are connected in series, KVL can be applied to obtain the total voltage. The combined voltage is the algebraic sum of the voltages of the individual sources. For example,

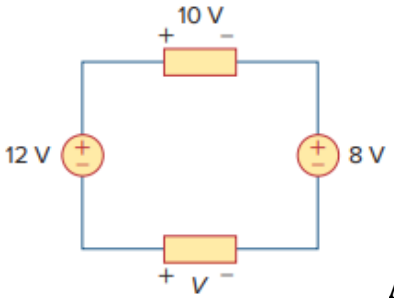


Vab = V1 + V2 − V3



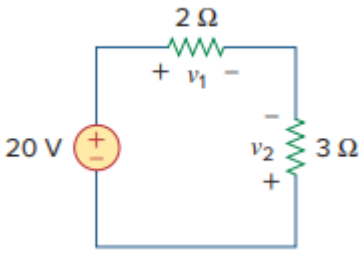
Exercise Qsn:

4. In the circuit in Fig., V is:



Ans: 12V-10V-8V+V=0 ⇒ V= 6V

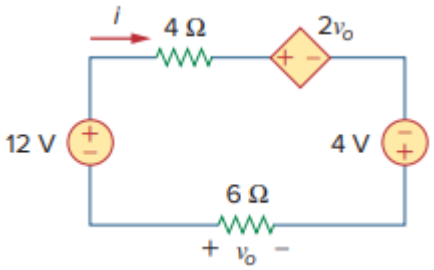
5. For the circuit in Fig. , find voltages v1 and v2



Ans: Applying KVL around the loop gives −20 + v1 − v2 = 0

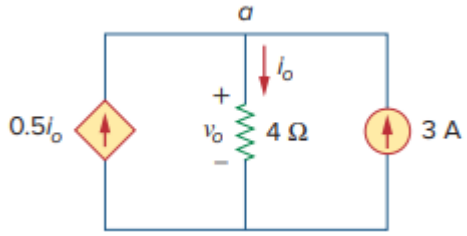
−20 + 2i + 3i = 0
or 5i = 20 ⇒ i = 4 A
v1 = 2i, v2 = −3i
Hence v1 = 8 V, v2 = −12 V

6. Determine Vo and i in the circuit shown



Ans: Applying KVL around the loop gives −12 + 4i + 2 vo − 4 + 6i = 0
Applying Ohm's law to the 6-Ω resistor gives vo = −6i
Hence , i = −8 A and Vo = 48 V

7. Find current i_o and voltage Vo in the circuit shown

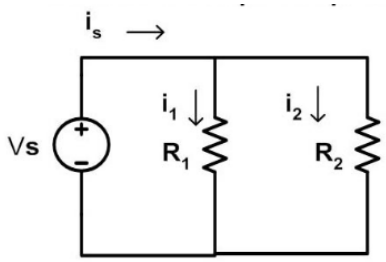


Ans: i_o = 6 A and Vo = 4i_o = 24 V

Current Division:

When a number of network elements are connected in parallel, the current divides through the elements. Therefore, in a parallel circuit, the current divider rule is used for the circuit analysis. And the voltage drop across all the parallel connected impedances remains the same.

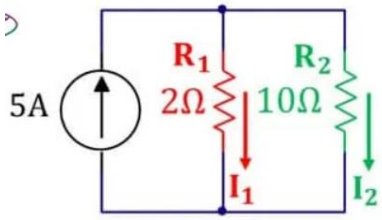
A parallel circuit with two resistors will divide the applied current Is into two currents $I_1$and I_2 through each resistor.



$$i_1 = \frac{R_2}{R_1 + R_2} i_s$$

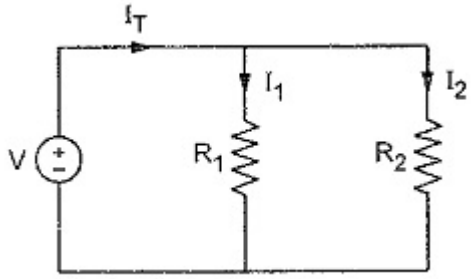
$$i_2 = \frac{R_1}{R_1 + R_2} i_s$$

Exercise1 : For the circuit shown, find the value of I_1 and I_2.



Ans: I_1=4.1675A and I_2= 0.8335A

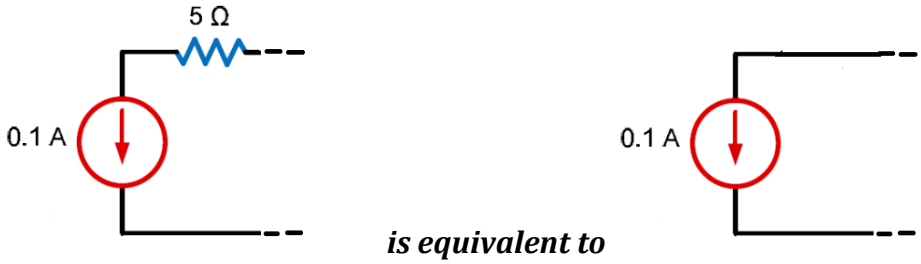
Exercise 2 : For the circuit shown, find the value of I_1 and I_2.



Given R1= 5KΩ and R2=10KΩ and Vs=12V

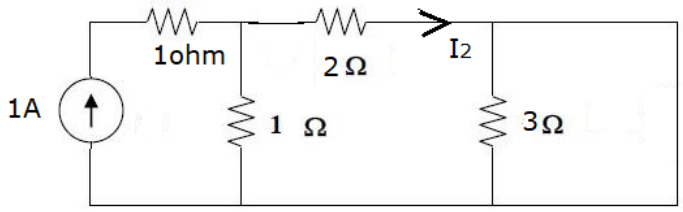
Ans: Req= 3.33KΩ , I_T=3.6mA, I_1=2.4mA and I_2= 1.2mA

Note: If any current source is in series with resistance then the resistance can be neglected.



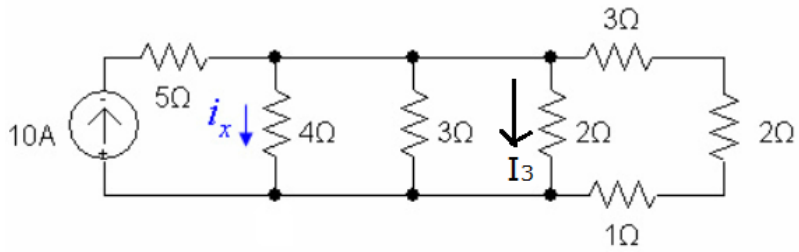
is equivalent to

Exercise 3 : For the circuit shown, find the value of I_2.



Ans: I_2= 1/3A

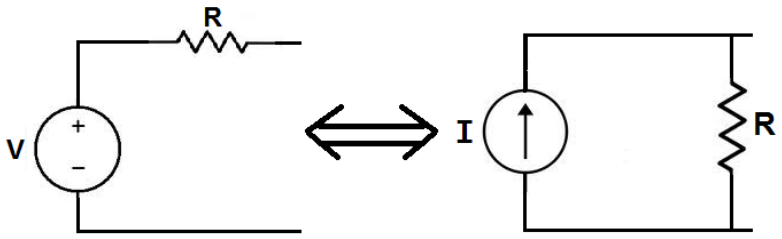
Exercise 4 : For the circuit shown, find the value of I_x and I_3.



Ans: ix=2A and I_3=3.99A

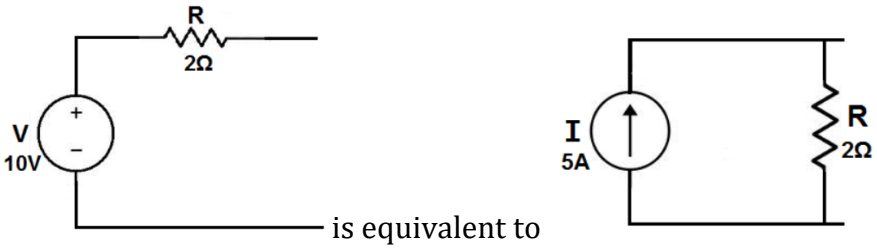
Note: Practically we can have a voltage source in series with resistance and current source in parallel with resistance

Source Transformation: *The source transformation of a circuit is the transformation of a power source from a voltage source to a current source, or a current source to a voltage source.*

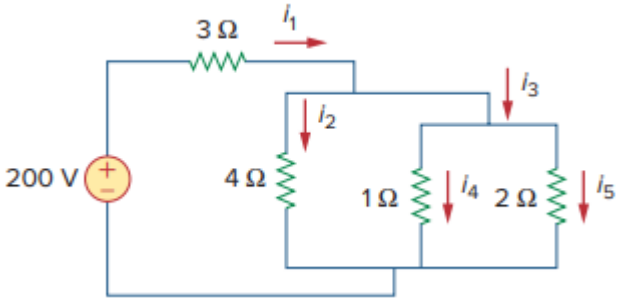


The transformation is done using ohm's law equation V=IR or I = V/R

Eg: A 10V voltage source with 2Ω series resistance can be transformed into 5A current source in parallel with 2 Ω resistance as shown in the figure below.



H.w Exercise 5 : For the circuit in Fig. determine i1 to i5.



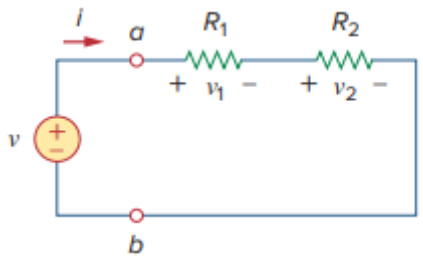
Answer: i1= 56A and i5=16A

Voltage Division:

In a circuit, when a number of elements are connected in series, input voltage divides across the elements. The current flowing through the series connected elements are always the same. And the amount of voltage drop across the resistor depends on its resistance value.

To calculate the voltage across each element, the voltage divider rule(potential divider rule) is used.

A series circuit with two resistors will divide the applied voltage V into two voltages V1 and V2 across each resistor.



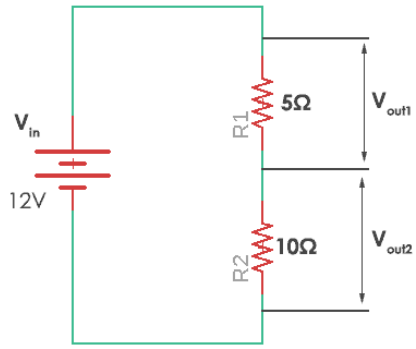
$$v_1 = \frac{R_1}{R_1 + R_2} v, \qquad v_2 = \frac{R_2}{R_1 + R_2} v$$

$V = V_1 + V_2$

In general, if a voltage divider has N resistors (R1, R2, … , RN) in series with the source voltage v, the nth resistor (Rn) will have a voltage drop of

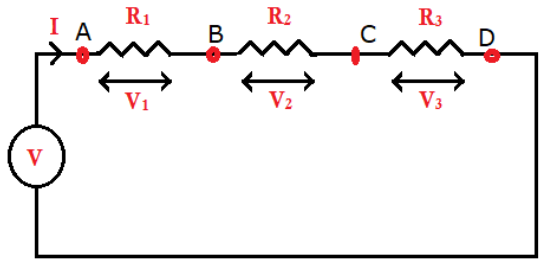
$$v_n = \frac{R_n}{R_1 + R_2 + \cdots + R_N} v$$

Exercise1 : For the circuit shown, find the value of Vout1 and Vout2.



Ans: Vout1 = 4V and Vout2 = 8V

Exercise 2 : For the circuit shown, measure the voltage between the terminals AB, BC, CD, AC, AD. Choose R1=4Ω, R2=8Ω, R3=12Ω and V=24V



Ans: V_{AB} = 4V, V_{BC}=8V, V_{CD}=12V, V_{AC}=12V and V_{AD} = 24V

Mesh Analysis:

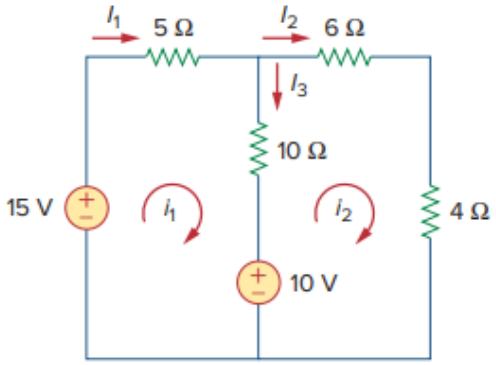
Mesh analysis provides another general procedure for analyzing circuits, using mesh currents as the circuit variables

Steps to Determine Mesh Currents:

1. Assign mesh currents $i_1, i_2, \ldots, i_n$ to the n meshes.
2. Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
3. Solve the resulting *n* simultaneous equations to get the mesh currents.

Sample Problems:

1. For the circuit in Fig. find the branch currents I_1, I_2, and I_3 using mesh analysis.



Solution:

We first obtain the mesh currents using KVL. For mesh 1

$$-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$$

or

$$3i_1 - 2i_2 = 1$$

For mesh 2,

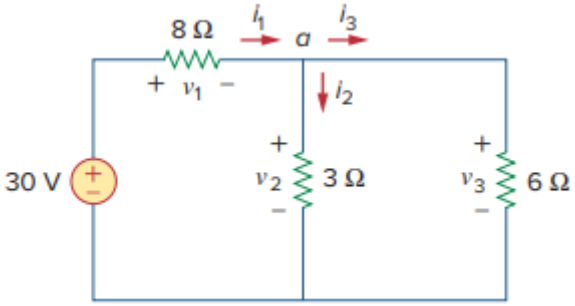
$$6i_2 + 4i_2 + 10(i_2 - i_1) - 10 = 0$$

or

$$i_1 = 2i_2 - 1$$

Solving the 2 mesh equations, we have I_1 =1 A, I_2 = 1 A, I_3 = 0

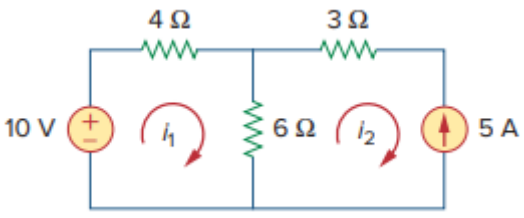
2. Find currents and voltages in the circuit



Ans: i1 = 3 A, i2 = 2 A, i3 = 1 A, v1 = 24 V, v2 = 6 V, v3 = 6 V

Mesh Analysis with Current Sources:

CASE 1: When a current source exists only in one mesh:

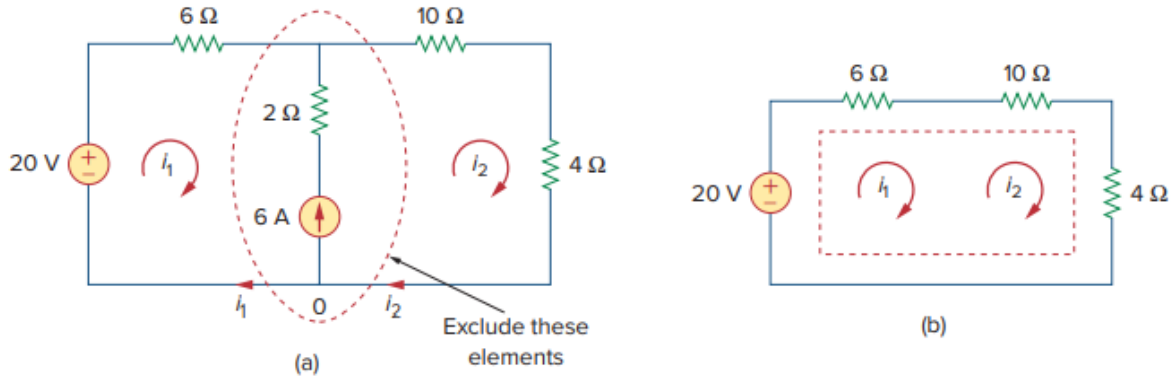


Here We set i2 = −5 A and write a mesh equation for the other mesh in the usual way; that is,

+10 - 4i1 - 6(i1 − i2) = 0

⇒ i1 = −2 A

CASE 2: When a current source exists between two meshes:



- ★ A **supermesh** results when two meshes have a (dependent or independent) current source in common.
- ★ A supermesh as the periphery of the two meshes and treat it differently.
- ★ If a circuit has two or more supermeshes that intersect, they should be combined to form a larger supermesh.
- ★ The current source in the supermesh provides the **constraint equation** obtained by applying KCL necessary to solve for the mesh currents.
- ★ Constraint equation gives the relationship between the individual mesh currents and the current source.
- ★ A supermesh has no current of its own. Hence KVL can be applied for the supermesh circuit.

Thus a supermesh requires the application of both KVL and KCL.

Therefore, apply KVL to supermesh shown in Fig (b) we have the equation;

+20 - 6i1 - 10i2 - 4i2 = 0

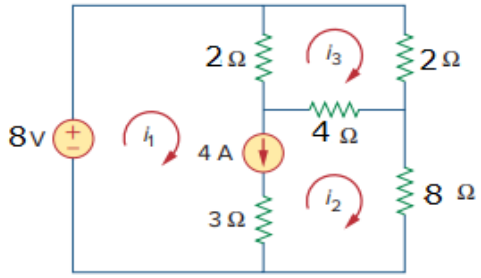
Applying KCL to node 0 in Fig.(a) gives i2 = i1 + 6

Solving these equations we have,

i1 = −3.2 A, i2 = 2.8 A

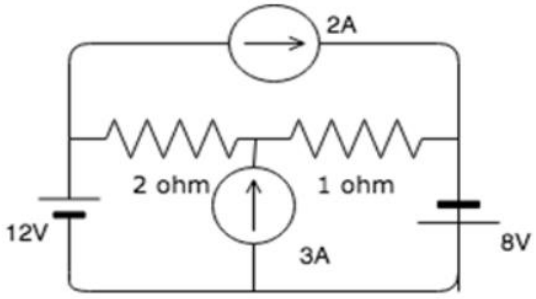
Exercise:

1. Use mesh analysis to determine i1, i2, and i3 in Fig.



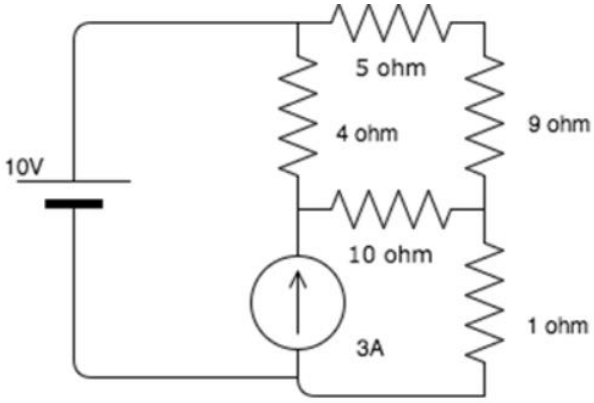
Answer: i1 = 4.632 A, i2 = 0.631 A, i3 = 1.4736 A

2. Find the mesh currents for the circuit shown:



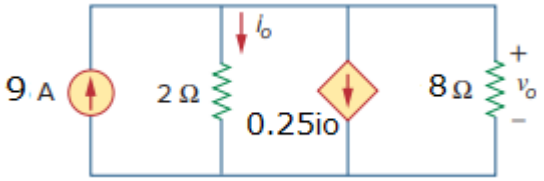
Answer: I1=7.67A, I2=10.67A and I3=2A

3. Find the mesh currents.



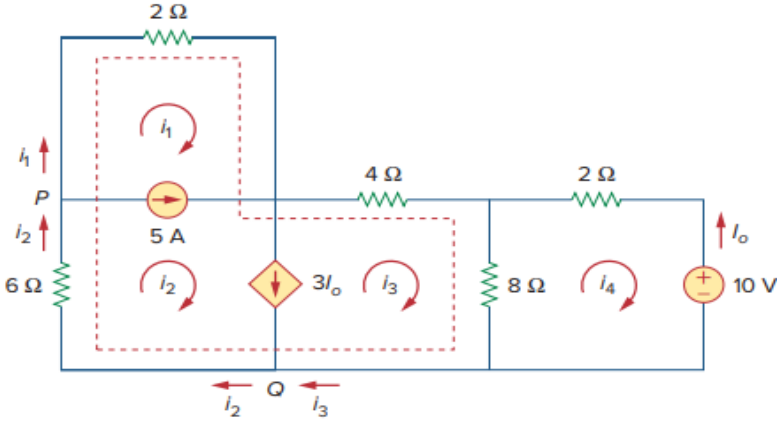
Answer:The currents are 2A, 1A, 0.57A.

4. Find vo and io in the circuit



Answer: 12 V, 6 A.

5. For the circuit in Fig. find i1 to i4 using mesh analysis.



Answer: i1 = −7.5 A, i2 = −2.5 A, i3 = 3.93 A, i4 = 2.143 A

Refer to problems solved in class*